

1. Static Testing & Dynamic Testing

From Transcript:

- **Static Testing:** Verifies documentation (requirements, design docs, HLD, LLD). Techniques: **Review, Walkthrough, Inspection.**
- **Dynamic Testing:** Tests actual software by executing code. Techniques: **Unit Testing, Integration Testing, System Testing, UAT.**

Additional Details (Online):

- Static testing helps catch defects early, reducing cost.
- Dynamic testing validates functional and non-functional aspects.

Interview Importance:

Frequently asked to check understanding of testing types.

Possible Interview Questions:

- What is the difference between static and dynamic testing?
- Give examples of static testing techniques.
- Why is static testing important?
- At which stage do we perform dynamic testing?

2. Review, Walkthrough & Inspection

From Transcript:

- **Review:** Formal or informal check of documents for correctness and completeness. Types: Requirement review, Design review, Code review, Test plan review.
- **Walkthrough:** Informal review led by the author; no formal meeting or minutes.
- **Inspection:** Formal review with roles (Reader, Writer, Moderator), scheduled meeting, documented minutes.

Additional Details:

- Reviews improve quality and reduce rework.
- Inspections are part of quality assurance processes in CMMI and ISO standards.

Interview Importance:

Common in QA interviews.

Possible Interview Questions:

- Explain review, walkthrough, and inspection with examples.
 - How is inspection different from walkthrough?
 - Who participates in an inspection meeting?
 - Why are these techniques called static testing?
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3. QA, QC & QE

From Transcript:

- **QA (Quality Assurance):** Process-oriented; ensures correct process is followed to prevent defects.
- **QC (Quality Control):** Product-oriented; actual testing to detect defects.
- **QE (Quality Engineering):** Advanced role; focuses on automation and engineering practices for quality.

Additional Details:

- QA is proactive; QC is reactive.
- QE involves coding skills for automation frameworks.

Interview Importance:

Very important for differentiating roles.

Possible Interview Questions:

- Difference between QA and QC?
 - What is QE and how is it different from QA/QC?
 - Which is preventive and which is detective?
 - Why is QE gaining importance in modern SDLC?
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4. Levels of Software Testing

From Transcript:

- **Unit Testing:** Tests individual modules; done by developers; white-box technique.

- **Integration Testing:** Tests data flow between modules; can be incremental (Top-down, Bottom-up, Sandwich) or non-incremental.
- **System Testing:** Tests complete application against requirements; includes functional and non-functional testing.
- **UAT (User Acceptance Testing):** Done by customers; includes Alpha and Beta testing.

Additional Details:

- System testing includes UI, functional, performance, security, and usability testing.
- UAT ensures business requirements are met before production.

Interview Importance:

Frequently asked in QA and developer interviews.

Possible Interview Questions:

- Explain different levels of testing.
- What is the difference between system testing and UAT?
- What are top-down and bottom-up approaches in integration testing?
- Why is system testing considered black-box testing?

5. Unit Testing

From Transcript:

- Tests smallest unit of code; white-box technique; done by developers.
- Techniques: Basis path testing, Control structure testing, Mutation testing.

Additional Details:

- Tools: JUnit, NUnit, TestNG.
- Helps catch defects early in development.

Interview Questions:

- What is unit testing?
- Which tools are used for unit testing?
- Why is unit testing important?

- Explain basis path testing.
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6. Integration Testing

From Transcript:

- Tests interaction between modules; ensures data flow is correct.
- Approaches: Incremental (Top-down, Bottom-up, Sandwich), Non-incremental.

Additional Details:

- Tools: JUnit for code-level, Selenium for UI-level integration.
- Incremental preferred over non-incremental due to easier defect isolation.

Interview Questions:

- What is integration testing?
 - Explain top-down vs bottom-up approach.
 - Why is non-incremental integration testing not preferred?
 - Give an example of integration testing in a real application.
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7. System Testing

From Transcript:

- Tests complete system against requirements; black-box technique.
- Includes UI testing, functional testing, non-functional testing, usability testing.

Additional Details:

- Non-functional testing includes performance, security, compatibility.
- Requires separate environments and specialized tools (LoadRunner, JMeter).

Interview Questions:

- What is system testing?
 - Difference between functional and non-functional testing.
 - What types of testing are done under system testing?
 - Why is system testing critical before UAT?
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8. UAT (User Acceptance Testing)

From Transcript:

- Last level of testing; done by customers in real or simulated environment.
- Types: Alpha (at developer site), Beta (at customer site).

Additional Details:

- Confirms business requirements before production.
- Often involves real-world scenarios.

Interview Questions:

- What is UAT?
- Difference between Alpha and Beta testing.
- Who performs UAT?
- Why is UAT important?